HEAVY HAUL SEMINAR • APRIL 23-24, 2026

CSX

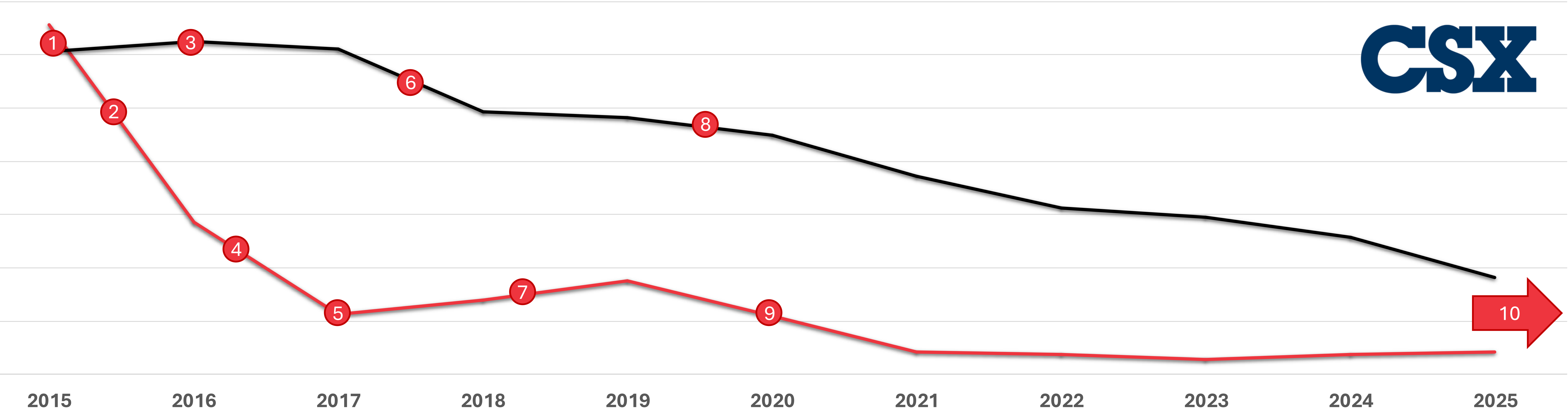
Refining the Rail Grinding Strategy to Improve Surface Conditions and Reduce Defects on CSX

DALLAS, TX

Dan Hampton
CSX

- Director of Contract Services
  - rail grinding
  - shoulder ballast cleaning
  - ballast distribution systems
  - ditching
  - undercutting
  - yard cleaning
  - vegetation management
- BS in Systems Engineering - West Point
  - Army Field Artillery School
  - Army Ranger School
- He is a member of AREMA Subcommittee 9 in Committee 4 – Rail.

Rail Grinding is Optimization



94%
Reduction in SSC / SDZ
2015 – 2025

83% Reduction
2015 – 2020
(Specialty Assets, Bridges, Curves)

62% Reduction
2021 – 2025
(Switch Points, RGT, Preventive)

70%
Reduction in TDD/TDC/TDT
2015 – 2025

26% Reduction
2015 – 2020
(Frequency & On Time)

60% Reduction
2021 – 2025
(Target Guage Corner RCF)

Technology and Process Improvements

1	Dan Hampton Takes Over Grinding Program	**6**	Planned Preventive Grinding (PPG) Crack Growth Schedule Grind
2	Mount Carbon Derailment – SSC masking a VSH	**7**	CSX Schedule Routing Software to Optimize Routing & General Depth of Cut Increase Led to Reduction in Track Miles
3	Combined RG400 & RGS Operations Increased depth of cut for all surface conditions	**8**	Increased Grinding Specifications of Gauge Corner Depth of Cut and Sensitivity of the Curve to Worst Case Conditions, Bridge RGS Program
4	Tonnage-Based Grind Cycles and Inspection-Based Grind Planning (Auto Skip tangents based on profile & surface conditions, and MGT since last grind)	**9**	Target Rail Conditions with RGS/RGT (Switch Point Patterns, Mid-Cycle Grind)
5	Bridge Grinding Policy based on substructure, SSC GPS Targeting Software	**10**	Implement New Optimized Target Profiles, Pattern Creation & More

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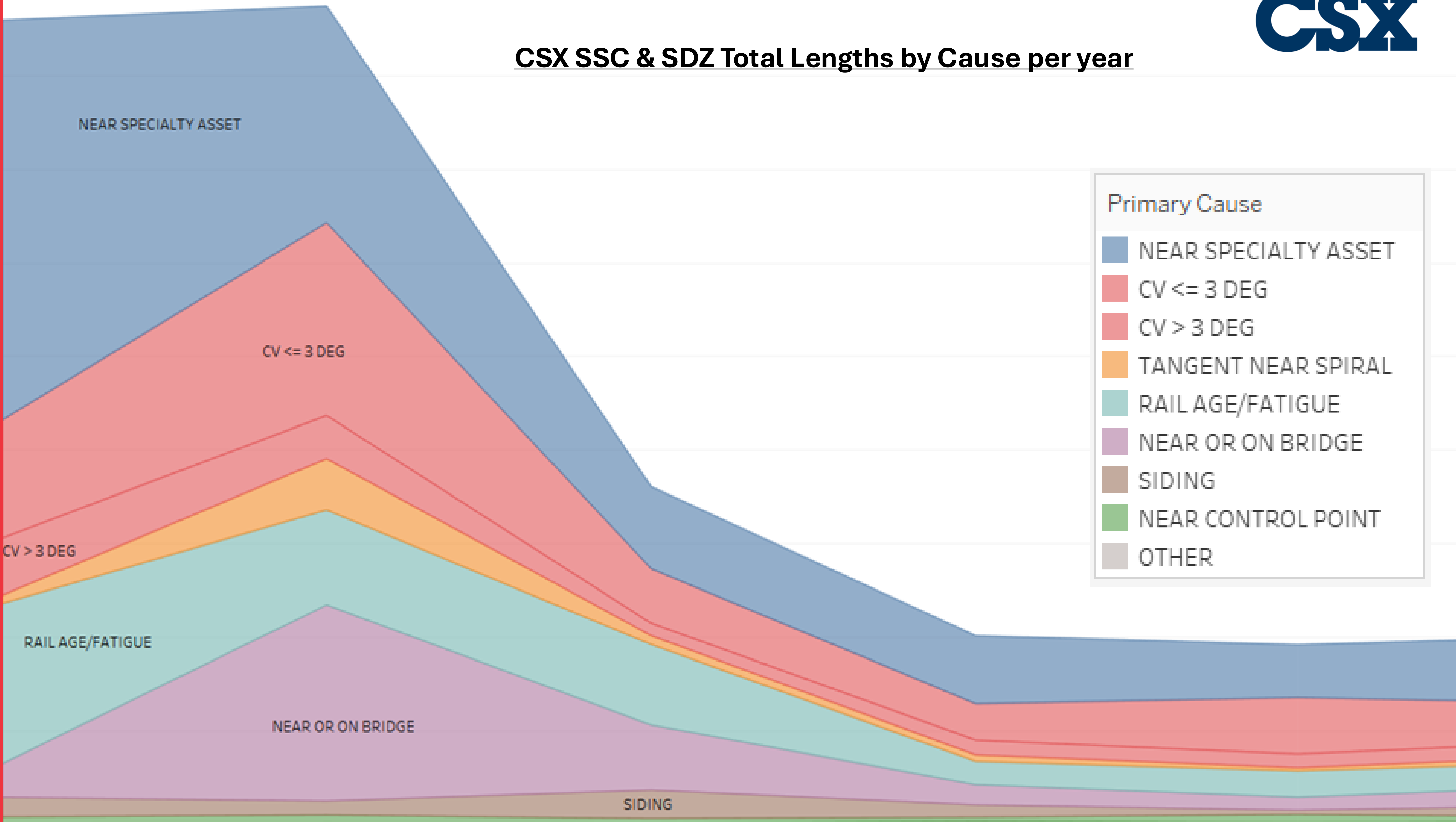
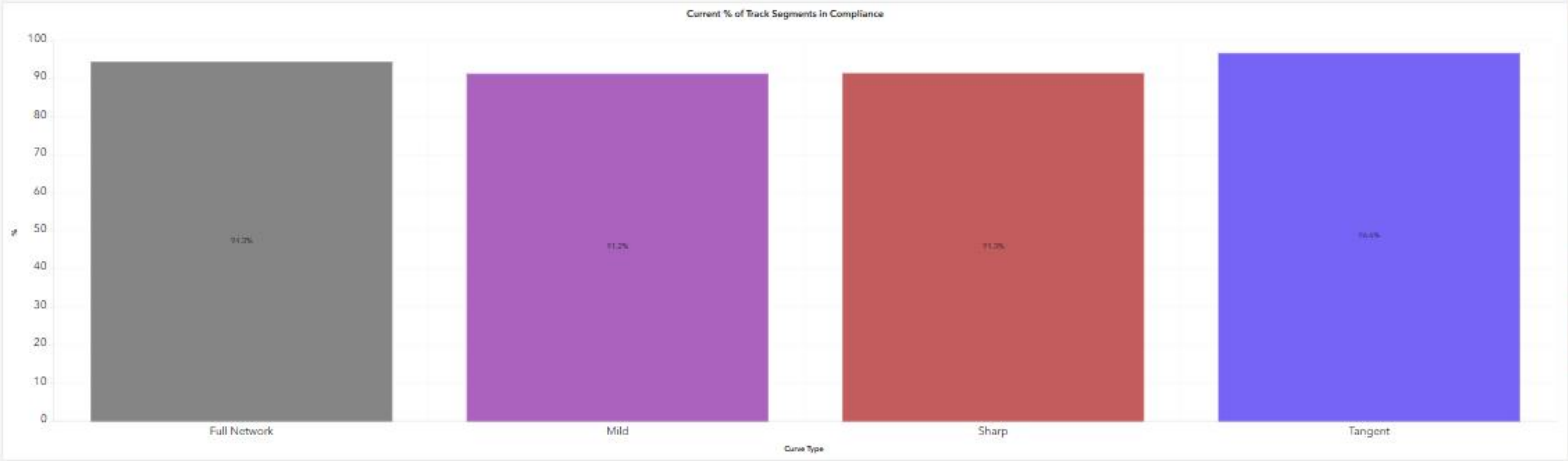
CSX SSC & SDZ Total Lengths by Cause per year



CSX - Platinum Asset Management Portfolio

COP | KPIs and Daily Stats | Scheduling Dashboard | Quality and Defects



Optimizing Scheduling

2015-2025 Update to Demand-Based Scheduling

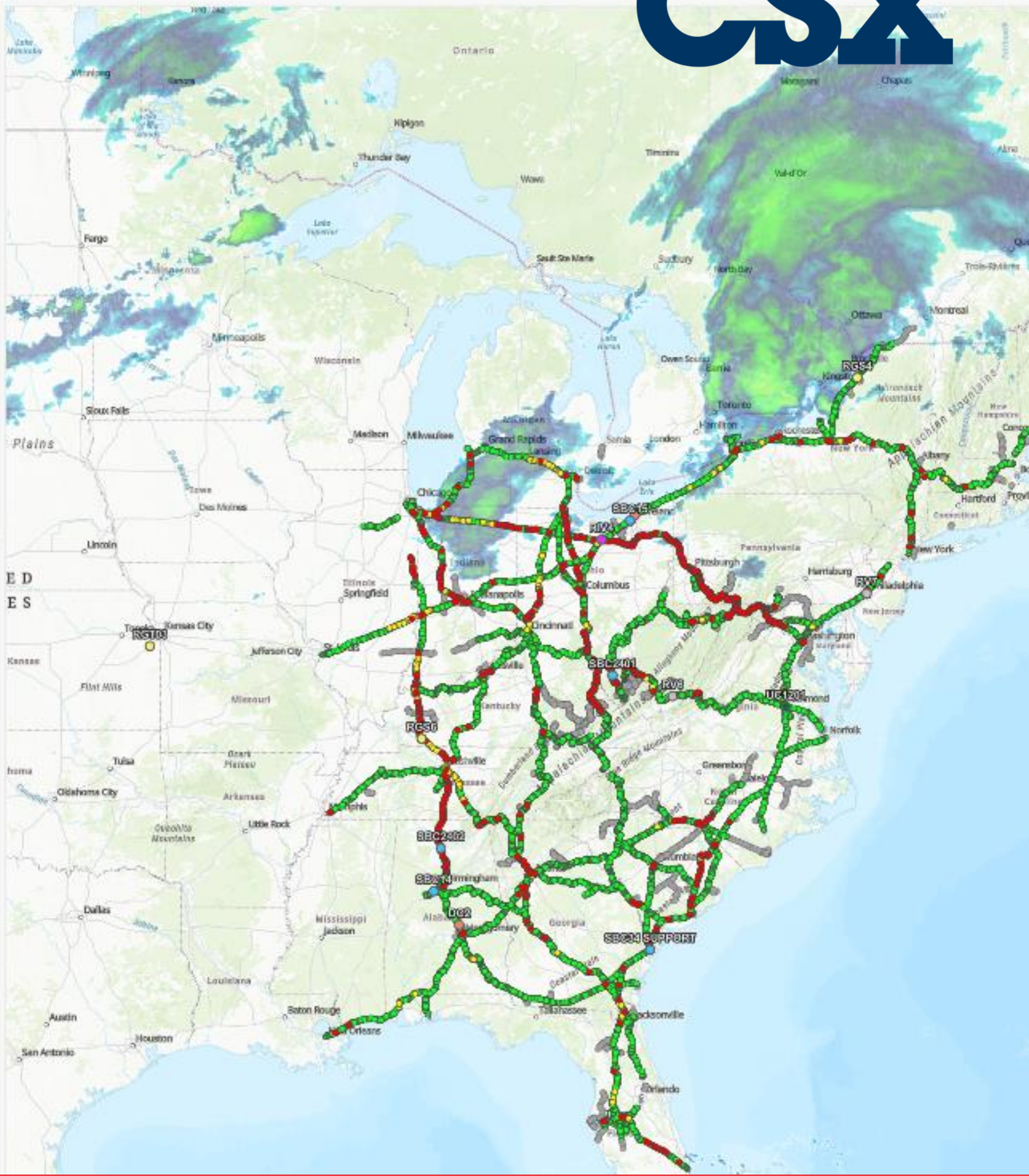
- Tonnage-Based Grind Cycles
- Planned Preventive Grinding (PPG) Damage Model
- CSX Schedule Routing Software to Optimize Routing
- Trade-Off of Grind Speed/Productivity vs. Depth of Cut

2026 – Updates to Condition-Based Scheduling

- Adding Grind History and Quality Inputs to Prioritize

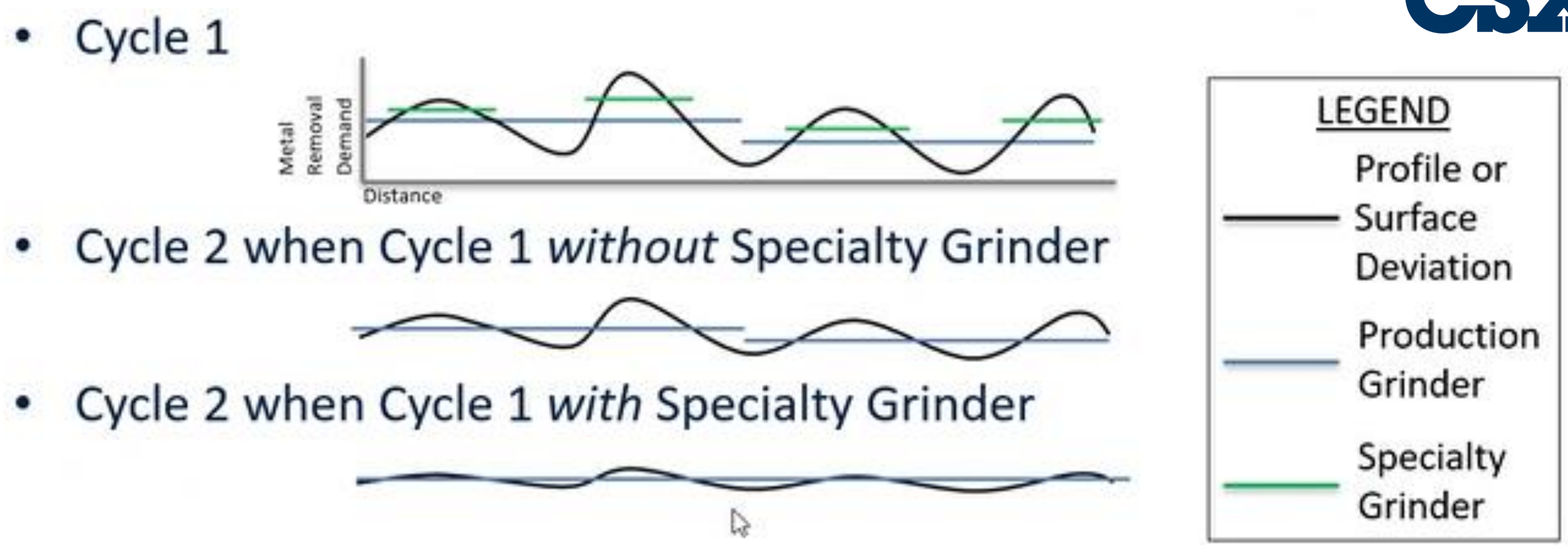
94%+

Network Compliance Before MGT Target



98% Arrival Compliance with 1.4h Additional Track Hours

Arrival	Track Distance % of Total	Arrival	Track Distance % of Total
Early	607 51%	Early	1,697 72%
On Time	205 17%	On Time	616 26%
Late	375 32%	Late	51 2%



Optimizing Execution

Grind Planning

- Inspection Based Grind Planning
- Trade-Off of Grind Speed/Productivity vs. DOC

Complementary Grind Plans

- Combined Operations and Grind Plans
- Target Rail Conditions

Vertical Split Head (VSH) Impact Analysis

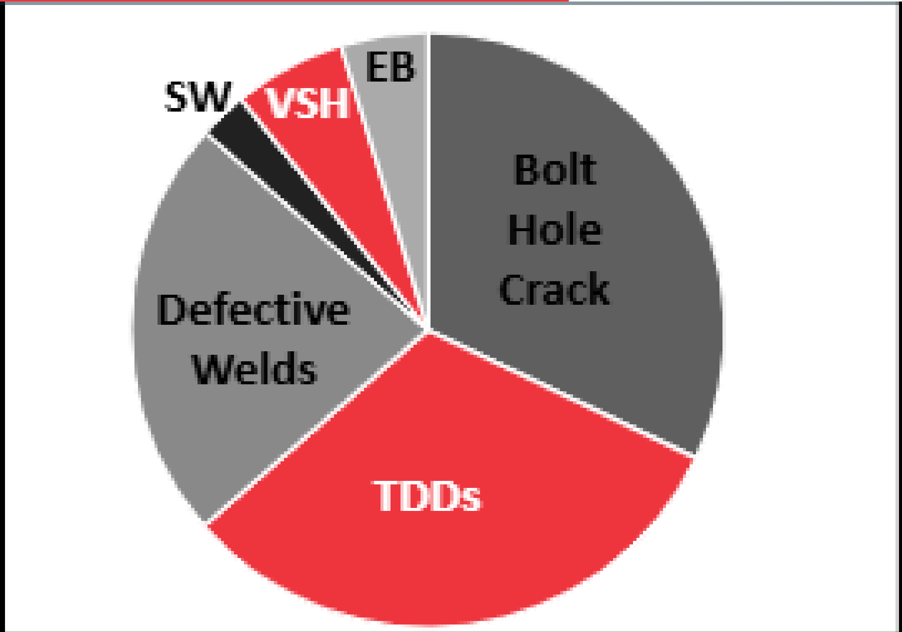
Low Frequency

High Risk

Lack of Signal Detection

6%

Total Class 1 Defects (WRI 2024)



5%

Rail Life <25 Years Old

300% Low Rails

66% Reduction in VSH 2021 – 2025

CSX

Wes Thomas
Loram

- Director of Digital Service Delivery
  - Rail Grinding Optimization
    - RailPro & NavPro (Execution)
    - PAMS (Scheduling)
    - Digital Twin (Rail Life & Economics)
  - Ballast cleaning optimization (RailDoctor)
  - Tie Capital Planning (Aurora)

- HBA in Business Admin - Western

- ICRI VTI Economics Group

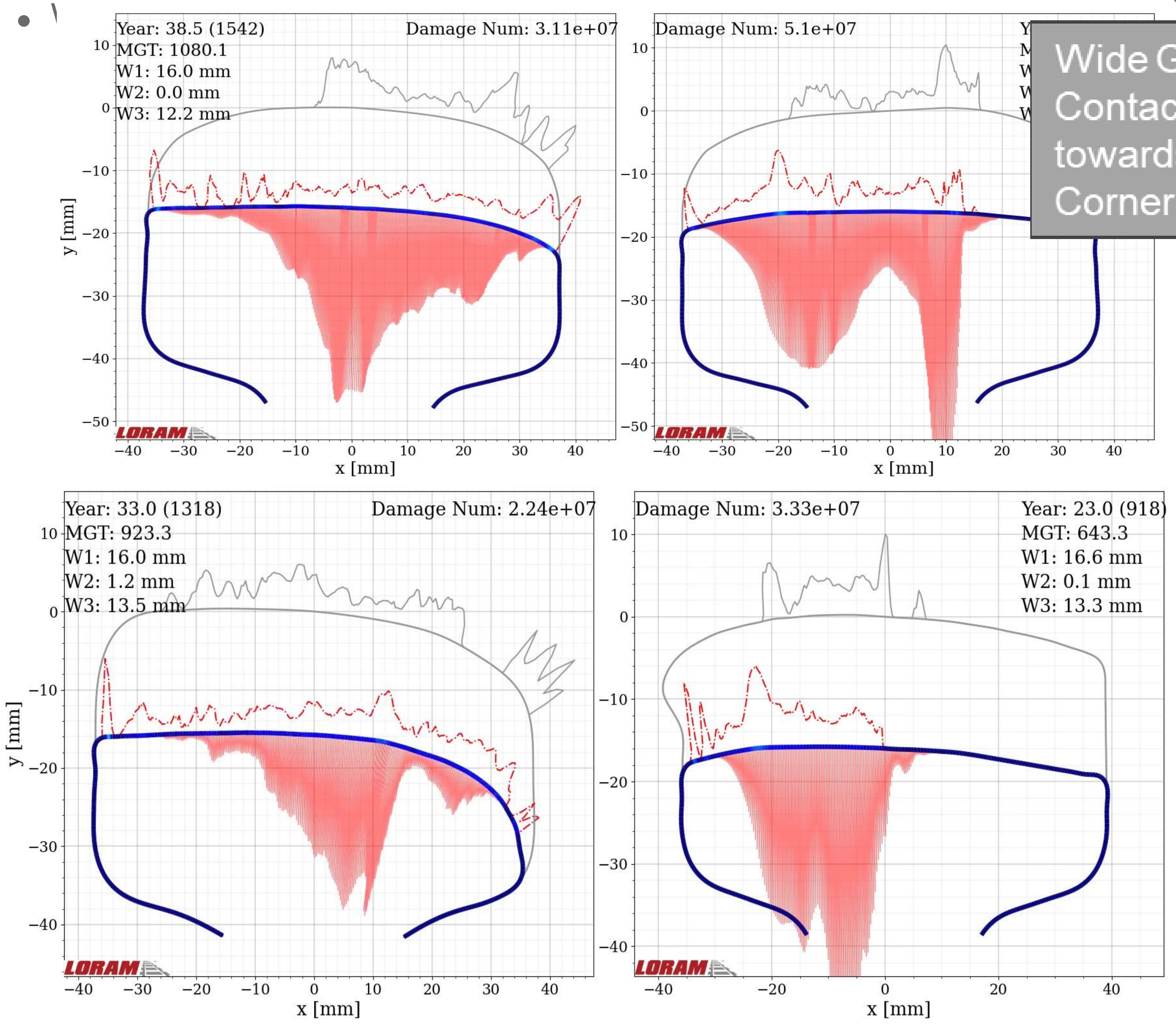
Vertical Split Head (VSH) Defects

- CSX has focused on preventing vertical split head defects (VSH) after 4+ broken rails in 2025. High risk

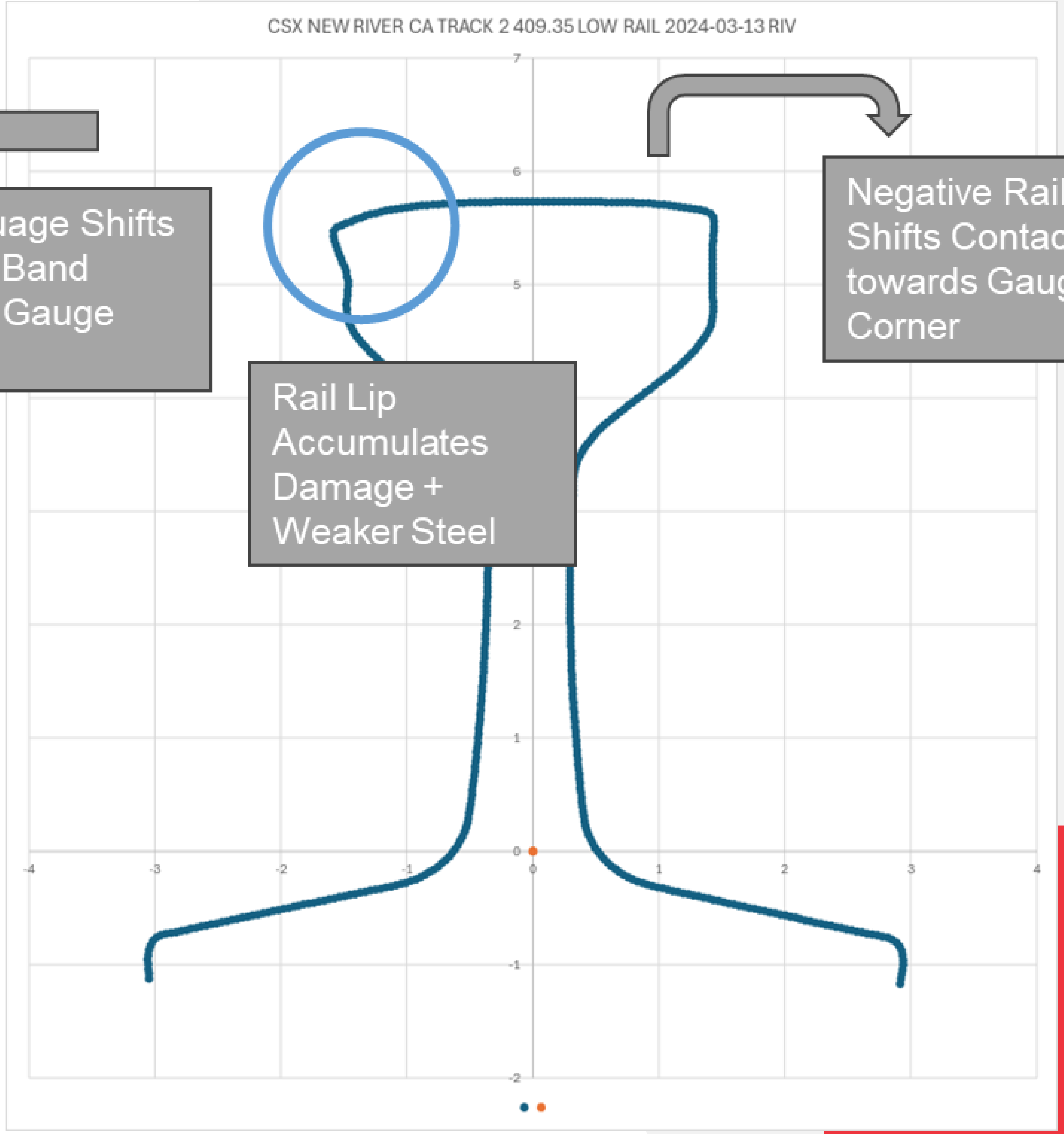
- Vertical Split Head (VSH) means a vertical split through or near the middle of the rail head
- A shear break is a longitudinal separation of the rail head, resulting from the loss of significant rail head parent metal and is not typically associated with inherent conditions in the material.
- A shear break usually occurs when the rail is loaded off the center axis, and can be associated with gaging problems, light weight rail, severely worn rail, or off-center loads caused by worn rolling stock wheels

FRA - Track Inspector Rail Defect Reference Manual - July 2015

Vertical Split Head (VSH) Root Cause Analysis



Wide Guage Shifts Contact Band towards Gauge Corner



Negative Rail Cant Shifts Contact Band towards Gauge Corner

Rail Lip Accumulates Damage + Weaker Steel

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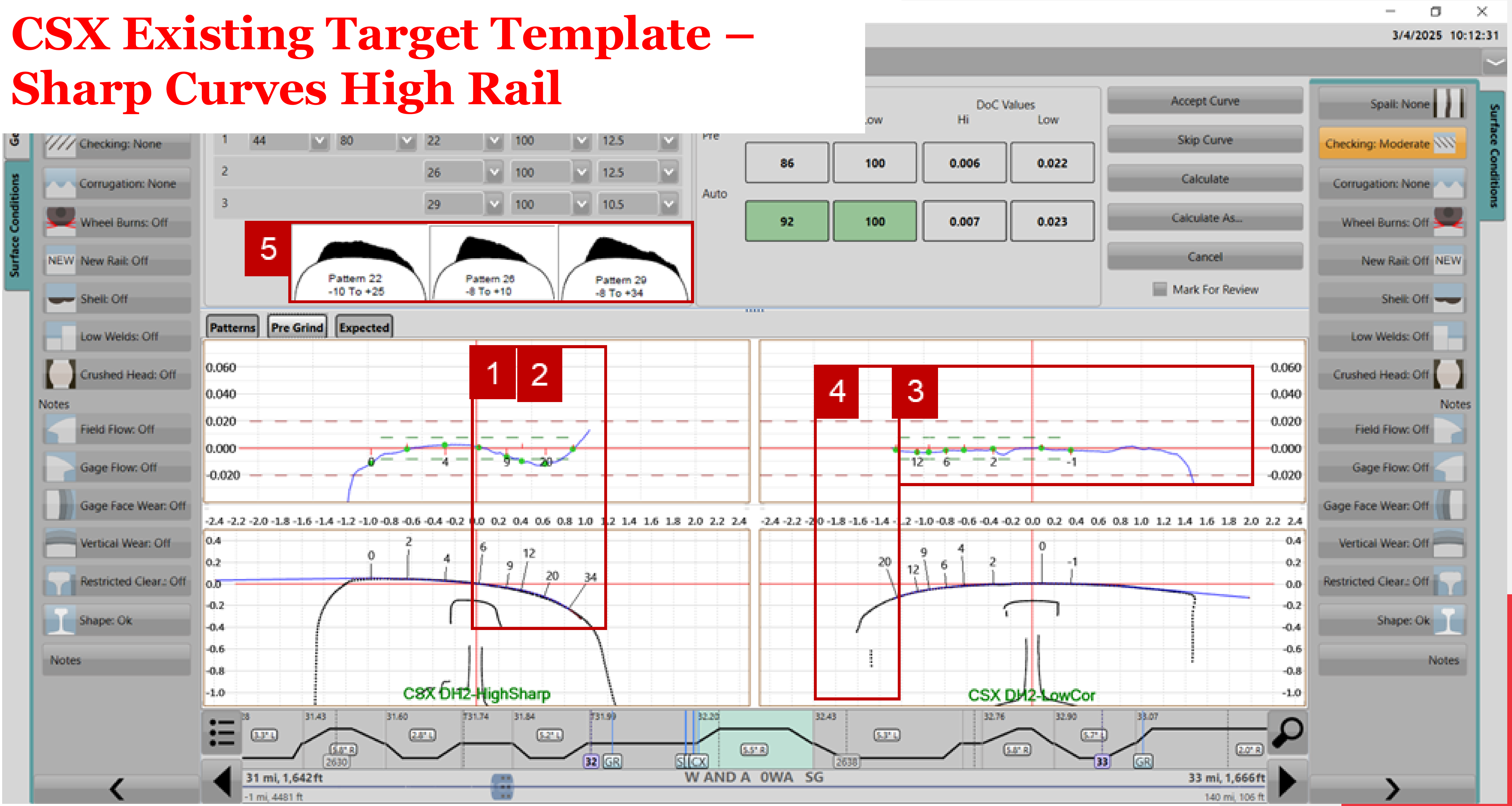
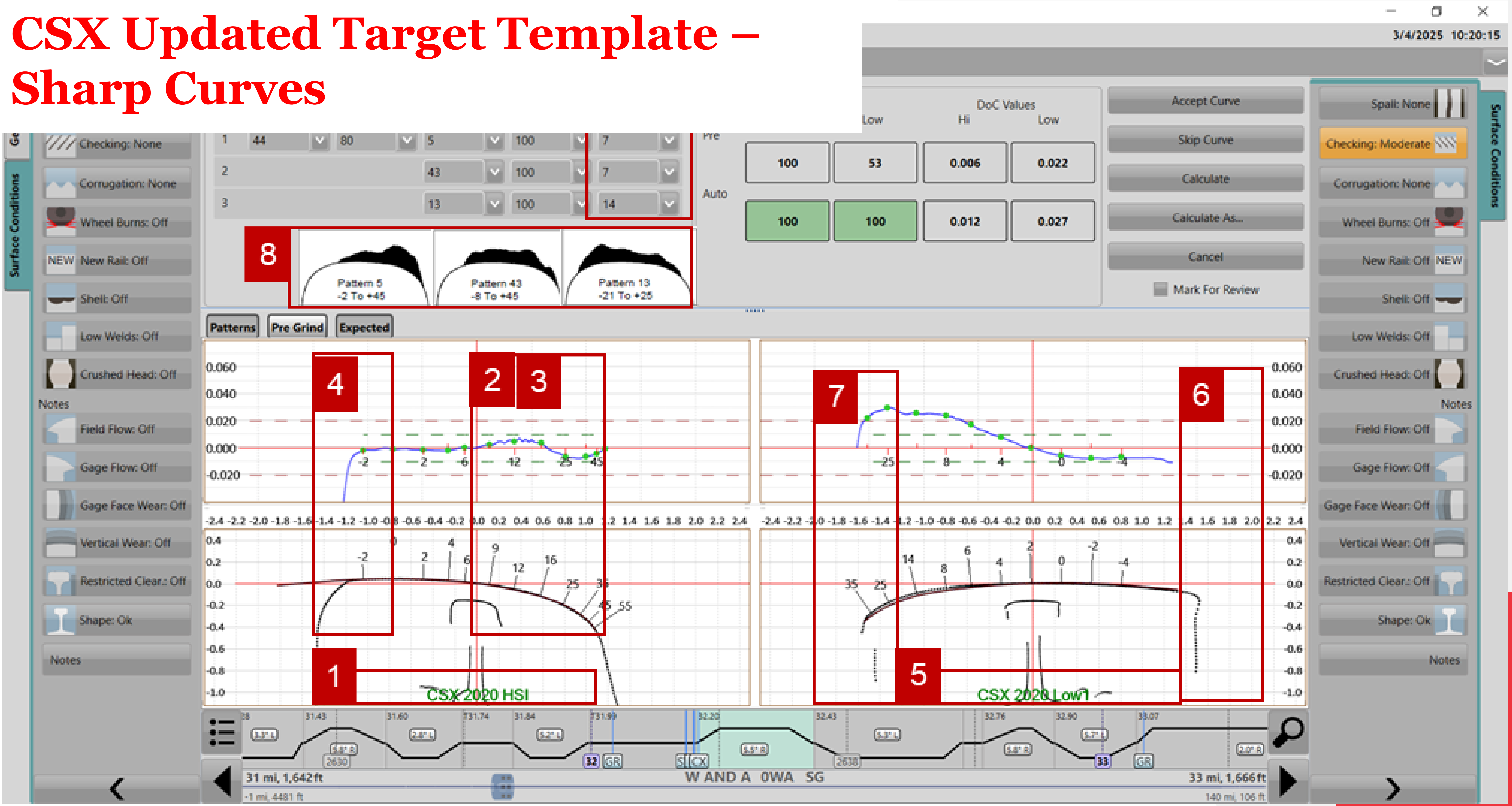
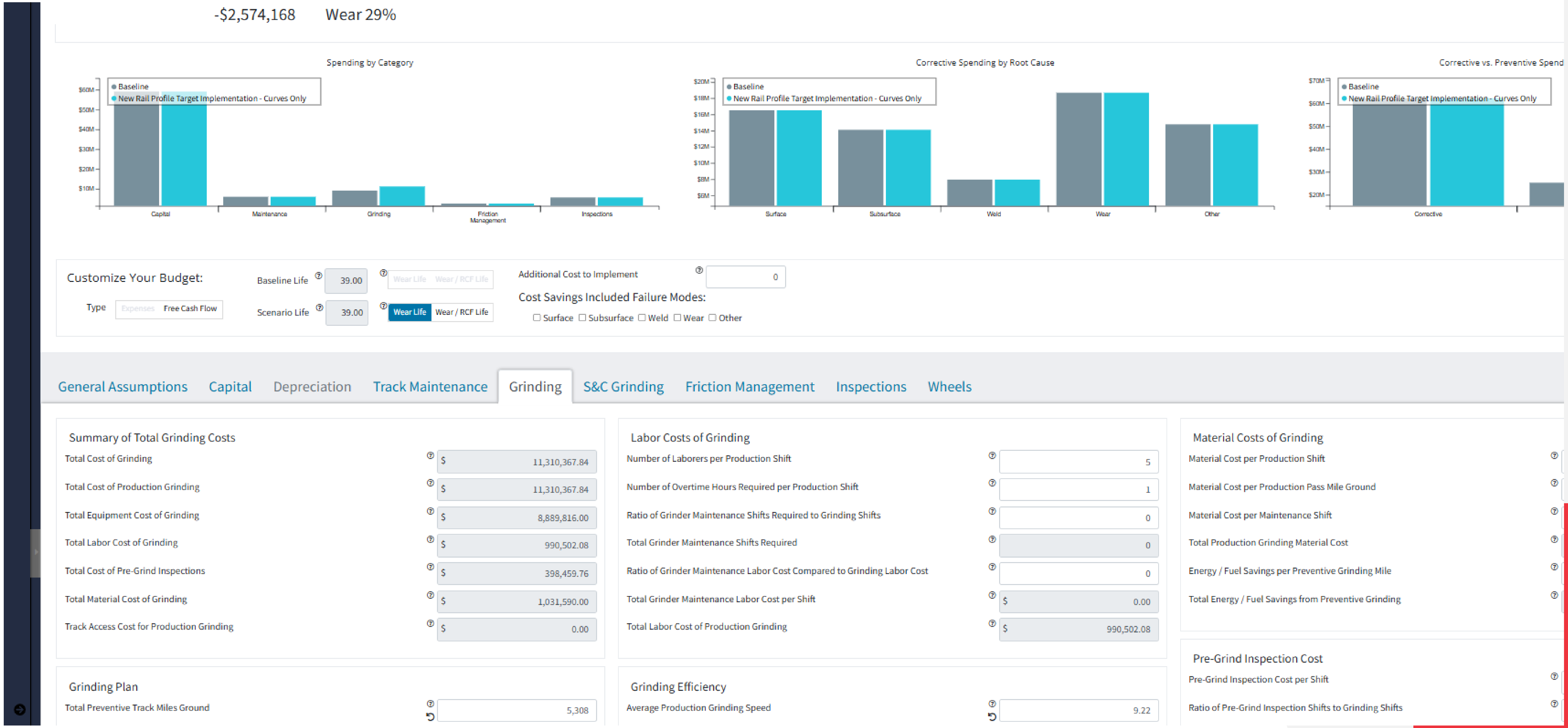
CSX Existing Target Template – Sharp Curves High Rail



CSX Updated Target Template – Sharp Curves



CSX Updated Target Template – Sharp Curves



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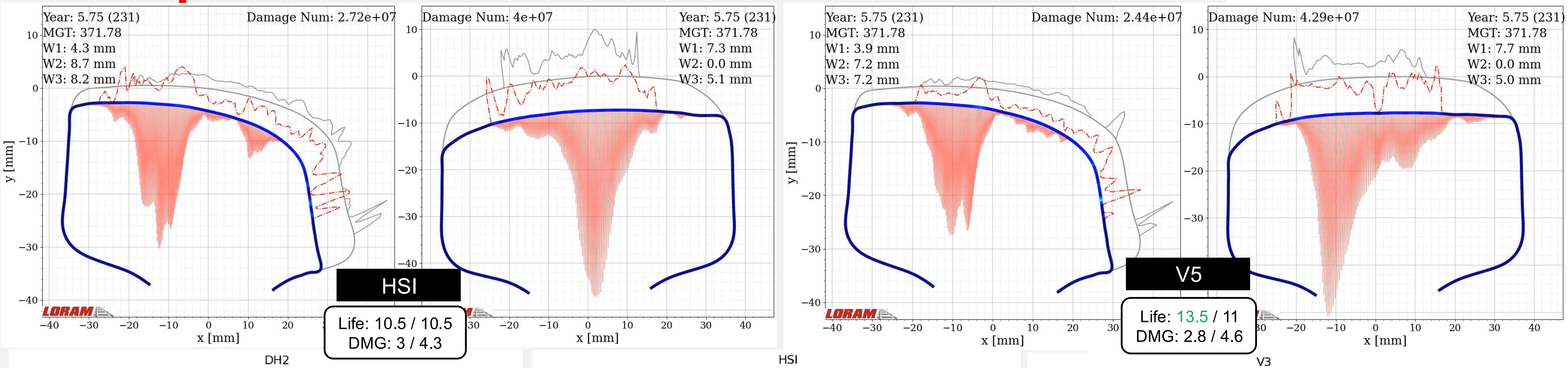
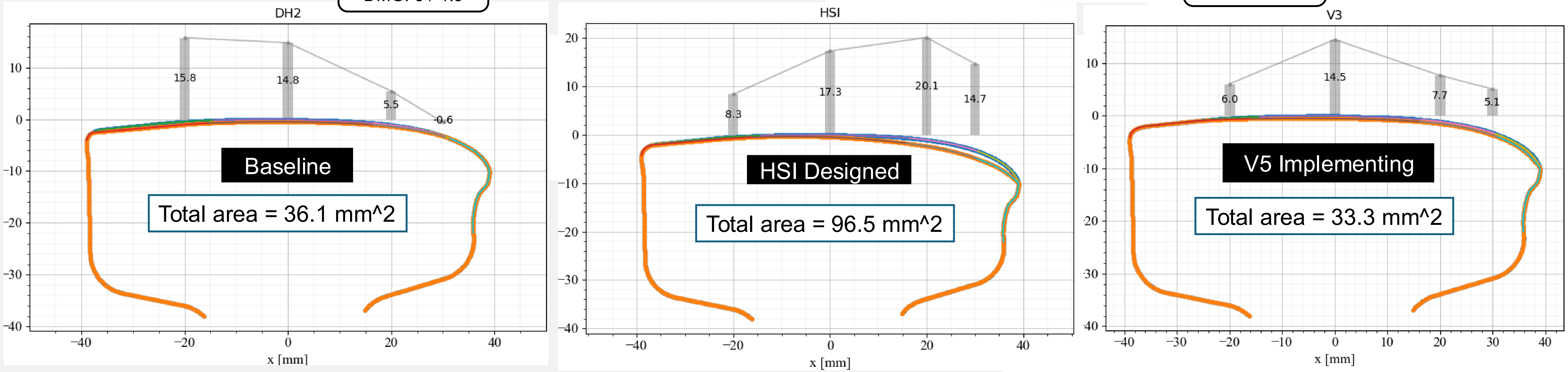
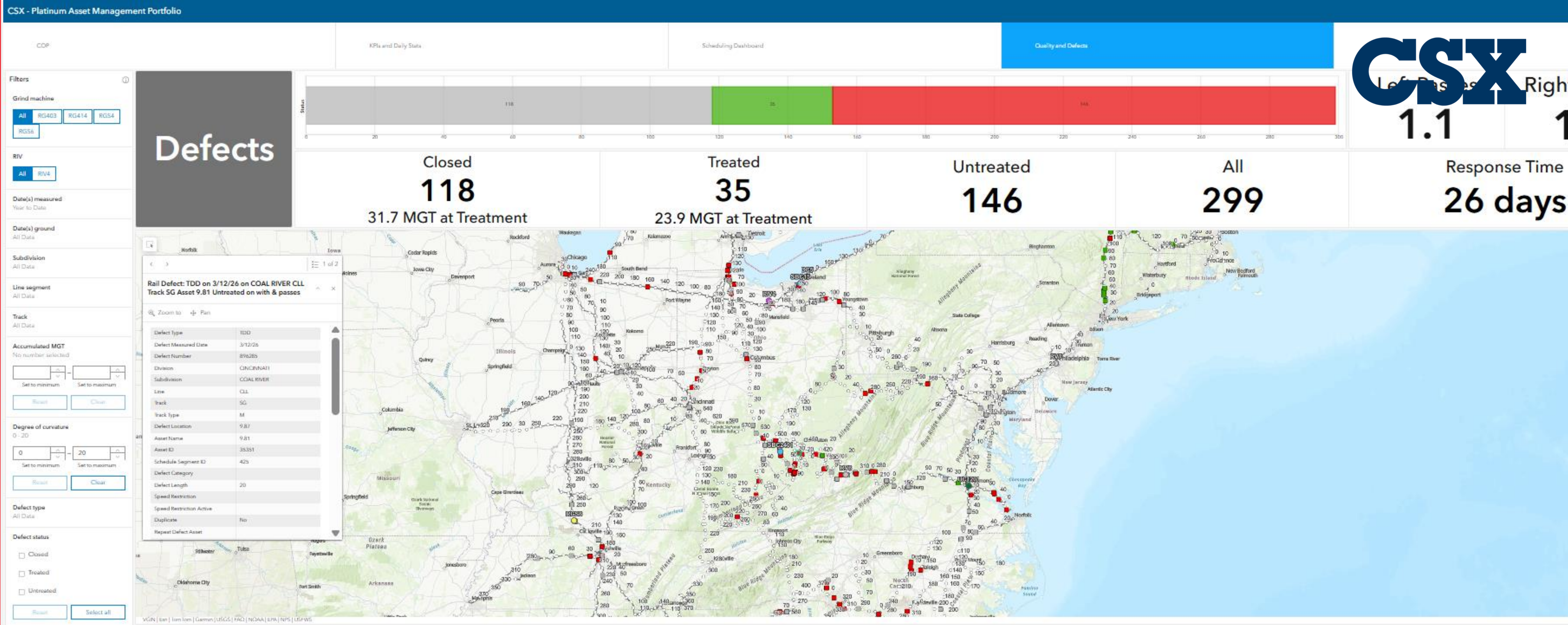
CSX Optimized* Target Template – Sharp Curves

*Wear Life Extension
Reduced RCF Risk
Targeted Gauge Corner Metal Removal
Optimized Productivity







Continuous Improvement

Defect Review Automation

- Tonnage Data Assigned to Defect
- Grind History – Root Cause Analysis
- Pre-Grind Inspection History – KPIs, Surface Condition, and Rail Profiles

Key Learnings

Optimization

- End of life field data alone takes decades to establish feedback loop and inability to isolate individual improvements
- Compare many different options to better understand the trade-offs of rail life, productivity, and economics

Data-Driven at Scale

- When possible, evaluate as many potential locations as possible to get the best sense of the root cause of the problem
- GQI is quality metric to measure how close post-grind profile matches target profile, not whether the target is right

Implementation

- Ensure the implementation is possible and does not lead to unforeseen challenges
- All components need to be evaluated, Wear Life, RCF Risk, Targeted Metal Removal, Productivity

Continuous Improvement

- Use field testing and wear & defect data trends to build confidence in the models for leadership and auditors
- Monitor developments in real-time to make adjustments to the program quickly